

Lab 6

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6B2

Sample code

```
#include <stdio.h>
#include <sys/wait.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <string.h>
int main()
{
    int shmid;
    char *shm;
    /* Create shared memory */

    shmid = shmget(IPC_PRIVATE, 1024, 0666 | IPC_CREAT);
    /* Attach shared memory */
    shm = (char *)shmat(shmid, NULL, 0);
    if (fork() == 0)
    {
        /* Child process reads */
        sleep(1);
        printf("Child read: %s\n", shm);
    }
    else
    {
        /* Parent process writes */
        strcpy(shm, "Hello from Parent");
        printf("Parent wrote to shared memory\n");
        wait(NULL);
        shmdt(shm);
        shmctl(shmid, IPC_RMID, NULL);
    }
    return 0;
}
```

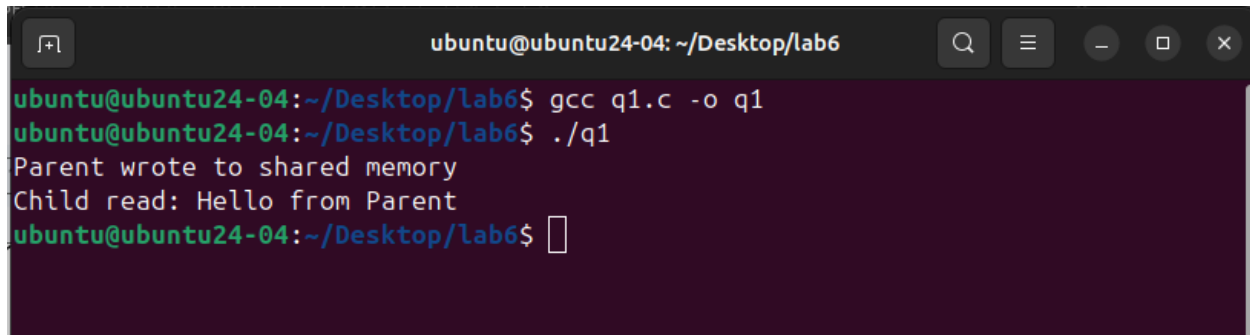
```
ubuntu@ubuntu24-04:~/Desktop/lab6$ gcc q1.c -o q1
ubuntu@ubuntu24-04:~/Desktop/lab6$ .q1
.q1: command not found
ubuntu@ubuntu24-04:~/Desktop/lab6$ ./q1
Parent wrote to shared memory
Child read: Hello from Parent
ubuntu@ubuntu24-04:~/Desktop/lab6$
```

Q1

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    int shmid;
    char *shm;
    shmid = shmget(IPC_PRIVATE, 1024, 0666 | IPC_CREAT);
    shm = (char *)shmat(shmid, NULL, 0);
    if (fork() == 0) {
        sleep(1);
        printf("Child read: %s\n", shm);
        shmdt(shm);
    }
    else {
        strcpy(shm, "Hello from Parent");
        printf("Parent wrote to shared memory\n");
        wait(NULL);
        shmdt(shm);
    }
}
```

```
    shmctl(shmid, IPC_RMID, NULL);  
}  
return 0;  
}
```

A terminal window titled 'ubuntu@ubuntu24-04: ~/Desktop/lab6' with search, menu, and window control icons. It shows the compilation and execution of a program. The user runs 'gcc q1.c -o q1', then './q1'. The output shows 'Parent wrote to shared memory' followed by 'Child read: Hello from Parent'.

```
ubuntu@ubuntu24-04:~/Desktop/lab6$ gcc q1.c -o q1  
ubuntu@ubuntu24-04:~/Desktop/lab6$ ./q1  
Parent wrote to shared memory  
Child read: Hello from Parent  
ubuntu@ubuntu24-04:~/Desktop/lab6$
```

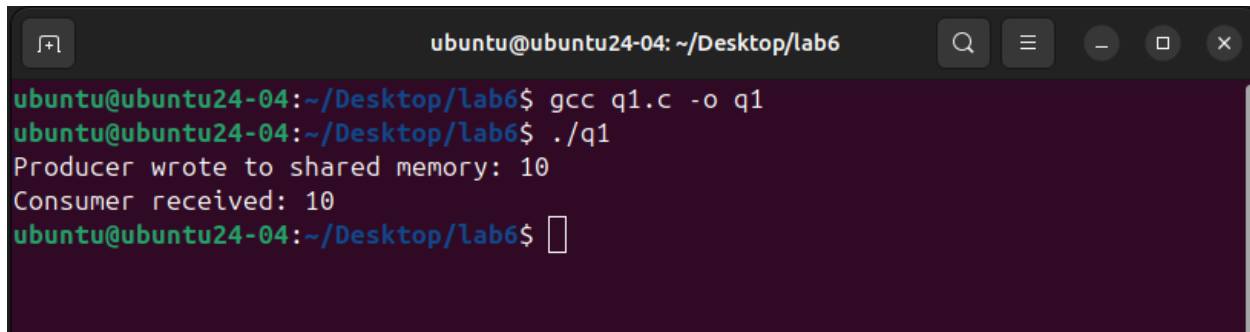
Q3

```
#include <stdio.h>  
#include <sys/ipc.h>  
#include <sys/shm.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main() {  
    int shmid;  
    shmid = shmget(IPC_PRIVATE, 1024, 0666 | IPC_CREAT);  
    if(fork()>0) {  
        int *shm= (int *)shmat(shmid, NULL, 0);  
        shm[0]=1;  
        shm[1]=10;    Write data to the shared memory  
        printf("Producer wrote to shared memory: %d\n", shm[1]);  
  
        shmdt(shm);  
    }  
}
```

```

else { // CHILD PROCESS
    int *shm= (int *)shmat(shmid, NULL, 0);
    while (shm[0] != 1);
    printf("Consumer received: %d\n", shm[1]);
    wait(NULL);
    shmdt(shm);
    shmctl(shmid, IPC_RMID, NULL);
}
return 0;
}

```



```

ubuntu@ubuntu24-04: ~/Desktop/lab6
ubuntu@ubuntu24-04:~/Desktop/lab6$ gcc q1.c -o q1
ubuntu@ubuntu24-04:~/Desktop/lab6$ ./q1
Producer wrote to shared memory: 10
Consumer received: 10
ubuntu@ubuntu24-04:~/Desktop/lab6$ 

```

Q2

Client

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>

```

```

int main()
{
    int shmid;
    char *shm;

    key_t key = 1234;

```

```

shmid = shmget(key, 1024, 0666);

shm = (char*) shmat(shmid, NULL, 0);

printf("Client received: %s", shm);

// convert lowercase to uppercase
for(int i=0; shm[i] != '\0'; i++)
{
    if(shm[i] >= 'a' && shm[i] <= 'z')
        shm[i] = shm[i] - 32;
}

printf("After converting to uppercase: %s", shm);

shmdt(shm); // detach

shmctl(shmid, IPC_RMID, NULL); // delete shared memory

return 0;
}

```

Server

```

#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main()
{
    int shmid;
    char *shm;

    key_t key = 1234; // shared key

    shmid = shmget(key, 1024, 0666 | IPC_CREAT);

    shm = (char*) shmat(shmid, NULL, 0);

    printf("Enter a string: ");

```

```
fgets(shm, 1024, stdin); // Reads from keyboard and stores it in shared memory
```

```
printf("Server wrote to shared memory.\n");
```

```
shmdt(shm); // detach
```

```
return 0;
```

```
}
```

A terminal window titled 'ubuntu@ubuntu24-04: ~/Desktop/lab6' with standard Ubuntu window controls. The terminal shows the compilation and execution of a C program using shared memory. The user runs 'gcc server.c -o server' and 'gcc client.c -o client'. Then they run './server', which prompts 'Enter a string: Hello' and outputs 'Server wrote to shared memory.'. Next, they run './client', which outputs 'Client received: Hello' and 'After converting to uppercase: HELLO'. The prompt returns to the user.

```
ubuntu@ubuntu24-04:~/Desktop/lab6$ gcc server.c -o server
ubuntu@ubuntu24-04:~/Desktop/lab6$ gcc client.c -o client
ubuntu@ubuntu24-04:~/Desktop/lab6$ ./server
Enter a string: Hello
Server wrote to shared memory.
ubuntu@ubuntu24-04:~/Desktop/lab6$ ./client
Client received: Hello
After converting to uppercase: HELLO
ubuntu@ubuntu24-04:~/Desktop/lab6$
```